10/10/2026

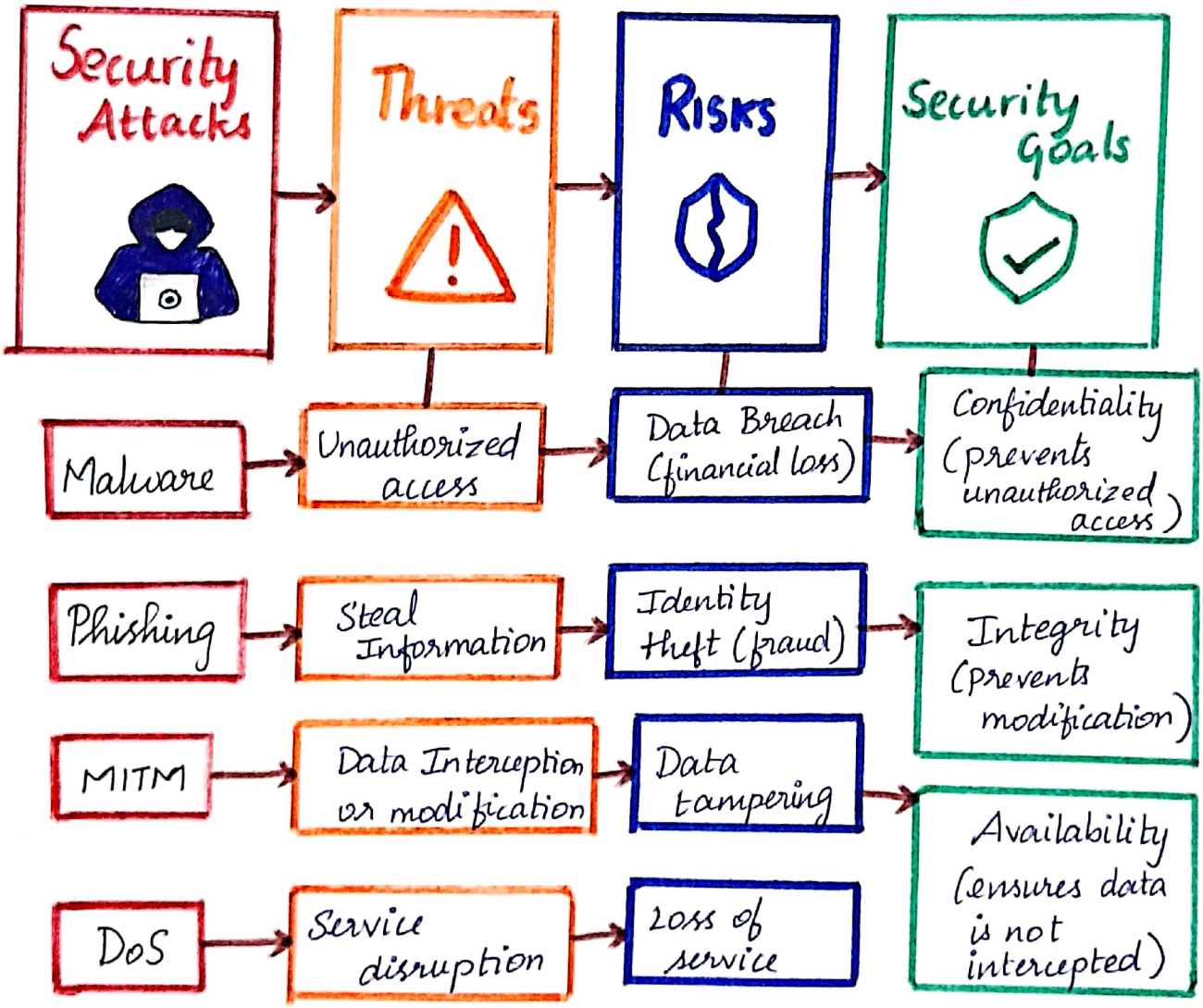
CSA5112 - CRYPTOGRAPHY & NETWORK SECURITY

CO1 - ASSESSMENT TOOL - 2

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1.

Security Attacks, Threats, Risks & Security Goals



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2.

SUBSTITUTION CIPHERS

Caesar Cipher



- Key Usage: Shift value (0-25)
- Complexity: Very low
- Security Strength: Weak

Monoalphabetic



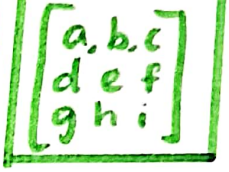
- Key Usage: 26-letter substitution
- Complexity: low
- Security Strength: Weak

Vigenere



- Key Usage: Keyword (multiple shifts)
- Complexity: Medium
- Security Strength: Moderate

Hill



- Key Usage: Key Matric (eg. 2x2, 3x3)
- Complexity: High
- Security Strength: Strongest

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3.

CLASSICAL ENCRYPTION SCHEMES

SUBSTITUTION

(replace characters)

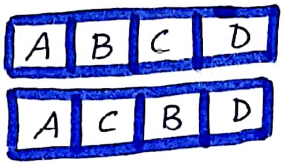
eg., Caesar, Vigenere, Hill



TRANSPOSITION

(rearrange characters)

eg., Rail Fence, Columnar, Row Transposition.



Data Structure	Changes the actual characters (same length)	Changes the order (same characters, same length)
Security level	Generally weaker	Slightly stronger (less affected by frequency)
Vulnerability	Easier to break (frequency analysis)	More resistant (but vulnerable to pattern analysis)

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4.

STEGANOGRAPHY

DATA HIDING



Hides secret data inside a cover file (image, audio, video, text)

Example :
Hidden text in an image

WATERMARKING



Adds ownership info to content

Example :
Digital watermark in photos/videos

SURVIVABILITY



Survives attacks / changes (noise, compression, etc.)

Example :
Robust watermarking in media

✓ Together, they help in secure communication by hiding, protecting & keeping the data safe

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